

Video Analysis Progress Tracking - Deployment Guide

Overview

This guide covers the deployment of the premium video analysis with progress tracking feature, which includes new database models, API endpoints, and achievements system.

✓ Pre-Deployment Checklist

1. Code Changes (Already Completed)

- [x] Created `VideoAnalysisProgress` model in Prisma schema
- [x] Created `VideoAnalysisAchievement` model in Prisma schema
- [x] Created `VideoAnalysisUserAchievement` junction table
- [x] Added `/api/video-analysis/progress` endpoint
- [x] Added `/api/video-analysis/achievements` endpoint
- [x] Created seed script for video achievements (`scripts/seed-video-achievements.ts`)
- [x] Updated `video-analysis-results.tsx` component
- [x] All TypeScript checks pass
- [x] Local build succeeds

2. Database Migration Required ⚠

The deployment is failing because the **production database** doesn't have the new tables yet. You need to apply the schema changes to your production database.

Option A: Using Prisma Migrate (Recommended)

```
cd /home/ubuntu/mindful_champion/nextjs_space

# Generate migration file
npx prisma migrate dev --name add_video_analysis_progress_and_achievements

# Apply to production (requires production DATABASE_URL)
DATABASE_URL="your_production_database_url" npx prisma migrate deploy
```

Option B: Using Prisma DB Push (Faster, but no migration history)

```
cd /home/ubuntu/mindful_champion/nextjs_space

# Push schema directly to production
DATABASE_URL="your_production_database_url" npx prisma db push
```

3. Seed Video Achievements

After applying the schema, seed the achievement definitions:

```
cd /home/ubuntu/mindful_champion/nextjs_space

# Run seed script
DATABASE_URL="your_production_database_url" npx ts-node scripts/seed-video-achievements.ts
```

4. Verify Environment Variables in Vercel

Ensure these environment variables are set in your Vercel project settings:

```
DATABASE_URL=your_production_postgres_url
NEXTAUTH_SECRET=your_secret_here
NEXTAUTH_URL=https://mindfulchampion.com
RESEND_API_KEY=your_resend_key
AWS_ACCESS_KEY_ID=your_aws_key (for video uploads)
AWS_SECRET_ACCESS_KEY=your_aws_secret
AWS_REGION=your_aws_region
AWS_S3_BUCKET_NAME=your_bucket_name
```

New Database Models

VideoAnalysisProgress

Tracks user progress for each skill type (serve, dink, drive, etc.):

- `userId` - Links to User model
- `skillType` - Type of shot being tracked
- `averageScore` - Running average of scores
- `totalShots` - Total shots analyzed for this skill
- `bestScore` - Personal best score
- `sessionsCount` - Number of analysis sessions
- `improvementRate` - Percentage improvement over time

VideoAnalysisAchievement

Defines available achievements:

- `code` - Unique identifier (e.g., "FIRST_ANALYSIS")
- `name` - Display name
- `description` - Achievement description
- `icon` - Icon name for UI
- `category` - Type: milestone, speed, score, accuracy, improvement, consistency
- `threshold` - Value needed to unlock
- `thresholdType` - Type: count, speed, score, percentage
- `points` - Reward points

VideoAnalysisUserAchievement

Junction table linking users to unlocked achievements:

- `userId` - Links to User
- `achievementId` - Links to Achievement
- `unlockedAt` - Timestamp when unlocked
- `value` - Actual value that triggered unlock (e.g., speed in MPH)

API Endpoints Added

GET /api/video-analysis/progress

Returns user's video analysis progress, history, and achievements:

```
{
  "stats": {
    "totalAnalyses": 15,
    "totalShots": 247,
    "avgScore": 78,
    "improvement": 12
  },
  "analyses": [...],
  "skillProgress": [...],
  "achievements": [...],
  "historyData": [...]
```

GET /api/video-analysis/achievements

Returns all achievements with user's unlock status:

```
{
  "achievements": [...],
  "byCategory": {
    "milestone": [...],
    "speed": [...],
    "score": [...]
  },
  "totalUnlocked": 5,
  "totalAchievements": 18,
  "totalPoints": 165
}
```

Deployment Steps

Step 1: Apply Database Changes

Choose one of the migration options above and apply it to your production database.

Step 2: Seed Achievements

Run the seed script to populate the 18 predefined achievements.

Step 3: Verify Build Locally

```
cd /home/ubuntu/mindful_champion/nextjs_space
npm run build
```

The build should complete successfully (it already does locally).

Step 4: Push to GitHub

```
cd /home/ubuntu/mindful_champion/nextjs_space
git add -A
git commit -m "Video analysis progress tracking - deployment ready"
git push origin master
```

Step 5: Trigger Vercel Deployment

Vercel will automatically deploy when you push to master. Monitor the deployment at:
<https://vercel.com/your-account/mindful-champion>

Step 6: Verify Deployment

After deployment succeeds:

1. Test Progress API:

```
bash
curl -H "Cookie: next-auth.session-token=YOUR_TOKEN" \
  https://mindfulchampion.com/api/video-analysis/progress
```

2. Test Achievements API:

```
bash
curl -H "Cookie: next-auth.session-token=YOUR_TOKEN" \
  https://mindfulchampion.com/api/video-analysis/achievements
```

3. Test UI:

- Upload a video at <https://mindfulchampion.com/train/video>
- Complete analysis
- Check for achievements and progress tracking



Troubleshooting

Issue: “Table ‘VideoAnalysisProgress’ does not exist”

Solution: You haven’t applied the database migration yet. Run the migration commands above.

Issue: “No achievements found”

Solution: Run the seed script to populate achievements.

Issue: Build fails with “Dynamic server usage” errors

Solution: These are warnings and don’t prevent deployment. The build completes successfully.

Issue: “Resend domain not verified”

Solution: This is a separate issue. See `RESEND_DOMAIN_SETUP_GUIDE.md` for email configuration.



Testing Achievements Locally

To test achievement unlocking logic locally:

```
# Start development server
npm run dev

# Upload a test video
# Complete analysis
# Check console logs for achievement triggers
# Verify achievements appear in UI
```

18 Available Achievements

Milestones:

- First Steps (1 analysis) - 25 pts
- Getting Serious (5) - 50 pts
- Dedicated Player (10) - 100 pts
- Analysis Pro (25) - 250 pts

Speed:

- Speed Demon (30 MPH) - 30 pts
- Power Player (40 MPH) - 50 pts
- Cannon Arm (50 MPH) - 100 pts

Score:

- Solid Technique (80+) - 40 pts
- Elite Form (90+) - 75 pts
- Pro Level (95+) - 150 pts

Accuracy:

- Perfect Shot (1 excellent) - 15 pts
- Hot Streak (5 excellent) - 50 pts
- Flawless Performance (all excellent) - 200 pts

Improvement:

- Getting Better (10% improvement) - 40 pts
- Major Progress (25% improvement) - 100 pts
- Transformation (50% improvement) - 250 pts

Consistency:

- Weekly Warrior (3 weeks streak) - 60 pts
- Monthly Dedication (4 videos/month) - 80 pts

What's New for Users

1. **Progress Tracking Dashboard** - View improvement over time with charts
2. **Skill-Specific Metrics** - Track progress for each shot type separately
3. **Achievement System** - Unlock 18 achievements across 6 categories
4. **Historical Analysis** - Compare current performance to past sessions
5. **Improvement Percentage** - See quantified improvement from first analysis

Post-Deployment Monitoring

After successful deployment, monitor:

1. **Error Logs** - Check Vercel logs for any API errors
2. **Database Performance** - Monitor query performance with new tables
3. **User Engagement** - Track achievement unlock rates
4. **API Response Times** - Ensure progress endpoint performs well

Commit Reference

The feature was implemented in commit: `ec8277f`

Changed Files:

- `prisma/schema.prisma` - Added 3 new models
- `app/api/video-analysis/progress/route.ts` - New endpoint
- `app/api/video-analysis/achievements/route.ts` - New endpoint
- `components/train/video-analysis-results.tsx` - UI updates
- `scripts/seed-video-achievements.ts` - New seed script

Success Criteria

Deployment is successful when:

- ☒ Build completes without errors
- ☒ Database has 3 new tables
- ☒ 18 achievements are seeded
- ☒ Progress API returns data
- ☒ Achievements API returns data
- ☒ UI displays progress and achievements
- ☒ Users can unlock achievements

Support

If you encounter issues:

1. Check Vercel deployment logs
2. Verify database connection
3. Ensure all environment variables are set
4. Check Prisma schema syntax
5. Verify seed script ran successfully

Last Updated: December 23, 2025

Feature Version: Premium Video Analysis with Progress Tracking v1.0